

SMARTTRACK AI LOGISTICS CONTROL TOWER

CARRIER PENALTY & DEMURRAGE RULEBOOK

PURPOSE: Official Carrier Penalty & Demurrage Rulebook for internal business rules reference and development team handoff for JSON translation and control tower system implementation.

SECTION 1 — CARRIER PENALTY & DEMURRAGE RULEBOOK (INDUSTRY REFERENCE DATA)

TRANSPORT MODE	CARRIER / REFERENCE	EQUIPMENT / SERVICE	FREE TIME	PENALTY AFTER FREE TIME	ESCALATION / SLA RULE
Ocean Freight – Dry	Maersk	20' Dry	4 days*	₹5,900/day	Escalate when free time is within 24 hours of expiry.
Ocean Freight – Dry	Maersk	40'/40HC Dry	4 days*	₹11,800/day	Higher rate applies in subsequent tariff slabs.
Ocean Freight – Reefer	Maersk	20' Reefer	4 days*	Carrier-specific	Immediate temperature and demurrage review.
Ocean Freight – Dry	MSC	20DV General	7 days	\$30/day initially	Rate increases in subsequent periods.
Ocean Freight – Dry	MSC	40DV General	7 days	\$60/day initially	Rate increases in subsequent periods.
Ocean Freight – Standard	DHL Global Forwarding	Standard container	4 working days	\$310/day Days 1–3; \$395/day Days 4–6; \$465/day Day 7+	Use the applicable tariff for the specific shipment.
Ocean Freight – Reefer	DHL Global Forwarding	Reefer container	2 working days	\$620/day Days 1–3; \$820/day Days 4–6; \$920/day Day 7+	Temperature and demurrage exposure require separate review.
Air / Express	DHL / FedEx	Express shipment	~24-hr operational target	Facility/airport-specific	Escalate if customer SLA is threatened.
Inland Freight	FedEx Freight	Trailer / freight equipment	Contract-specific	\$273/day without power*	Detention depends on equipment/service.

SECTION 2 — SMARTTRACK OPERATIONAL PENALTY RULES (SYSTEM RULES)

RULE ID	CONDITION	THRESHOLD	SYSTEM ACTION	ESCALATION
DEM-001	Ocean container approaching free-time expiry	≤24 hr remaining	Generate demurrage warning notification	Dispatcher
DEM-002	Free time expired	Day 1 after free time	Start demurrage cost calculation	Logistics Manager
DEM-003	Demurrage exposure increasing	>\$250/day	Display financial-risk alert	Logistics Manager
DEM-004	Extended port delay	>48 hr delay	Evaluate alternate port / feeder reroute	Supply Chain Manager
DEM-005	Critical cargo delay	>72 hr or inventory risk	Evaluate partial air expedite	Regional Director
DEM-006	Major financial exposure	>\$25,000	Require management approval	Regional Director
DEM-007	Full mode conversion	Ocean → Air	Executive approval required	VP Operations

RULE ID	CONDITION	THRESHOLD	SYSTEM ACTION	ESCALATION
DEM-008	Reefer temperature excursion	Outside product specification	Quarantine + QA assessment	Quality / Operations
DEM-009	Severe weather	Significant route/safety risk	Recalculate ETA and monitor carrier advisory	Control Tower
DEM-010	Port congestion	ETA deterioration >48 hr	Compare rerouting vs waiting cost	Logistics Manager

SECTION 3 — SMARTTRACK PLANNING BENCHMARKS

PARAMETER	SMARTTRACK ASSUMPTION
Standard ocean free time	3–5 days
Standard dry-container penalty	\$150–350/day/container
Reefer penalty planning range	\$300–900/day/container
Free-time warning	24 hours before expiry
Rerouting trigger	>48 hours delay
Critical escalation	>72 hours / inventory risk
Partial air expedite	25% of critical cargo
Full ocean → air conversion	VP approval

SECTION 4 — DEVELOPER HANDOFF — CORE BUSINESS LOGIC

```
IF free_time_remaining <= 24 hours
  → DEMURRAGE_WARNING

IF free_time_expired
  → START_DEMURRAGE_CALCULATION

IF vessel_delay > 48 hours
  → REROUTING_ASSESSMENT

IF vessel_delay > 72 hours OR critical_inventory_risk
  → AIR_EXPEDITE_ASSESSMENT

IF mitigation_cost > approval_limit
  → ESCALATE_TO_NEXT_AUTHORITY
```

SECTION 5 — IMPORTANT ASSUMPTIONS & DISCLAIMERS

IMPORTANT NOTICE & TARIFF DISCLAIMER

Carrier tariffs are lane-, port-, equipment-, booking- and contract-specific. Published carrier examples are used as industry references and benchmarks. Actual demurrage, detention, storage and handling charges must be confirmed against the applicable carrier tariff or contract.

SmartTrack penalty values are internal planning benchmarks and must NOT be represented as universal Maersk, MSC, DHL or FedEx policies.

The 48-hour rerouting trigger, 72-hour critical escalation threshold, \$250/day financial alert, \$25,000 escalation threshold and approval rules are proposed SmartTrack control-tower business rules.

SECTION 6 — REFERENCES

- **Maersk:** Ocean freight / demurrage and free-time references
- **MSC:** Ocean freight / detention and demurrage references
- **DHL Global Forwarding:** Ocean demurrage/detention tariff references
- **DHL Express:** Air/express service references
- **FedEx Freight:** Freight detention/tariff references
- **Hapag-Lloyd:** Ocean freight operational reference
- **Flexport:** Ocean disruption and rerouting reference